

AWS Cloud Infrastructure Report

Cloud & DevOps Internship — Task 2

AWS Cloud Infrastructure Provisioning with AWS – IAM, VPC, EC2, S3 & Secure Networking

Region: ap-south-1 — Asia Pacific (Mumbai)

1. Project Overview

This report documents the AWS infrastructure provisioned for Task 2 of the Cloud & DevOps Internship, covering IAM, VPC networking, EC2, Nginx, S3, CloudWatch, security controls, architecture documentation, and cost estimation.

2. IAM and Account Security

Configured IAM users are **DevOpsAdmin** and **CloudIntern**. Both use the **DevOpsAdmins** group for the internship laboratory, with **AdministratorAccess** attached to the group as specified by the task instructions. Root and IAM-user MFA evidence is included. The IAM Roles page also shows **EC2-S3-Access** with EC2 as the trusted service.

3. VPC and Networking

VPC: **cloud-devops-vpc** — 10.0.0.0/16. Public subnet: **public-subnet** — 10.0.1.0/24. Private subnet: **private-subnet** — 10.0.2.0/24 in ap-south-1a, with public IPv4 auto-assignment disabled. Internet Gateway: **cloud-devops-igw**. Public and private route tables are documented and evidenced.

4. EC2 and Nginx

EC2 instance: **cloud-devops-web-server**, Ubuntu, t3.micro, public subnet, Security Group **web-server-sg**, key pair **cloud-devops-key**. SSH and Nginx verification evidence is included.

5. S3 Storage

S3 bucket: **muzammil-cloud-devops-task2-2026**, Region ap-south-1, versioning enabled. The Task 2 test object was uploaded and verified.

6. CloudWatch Monitoring

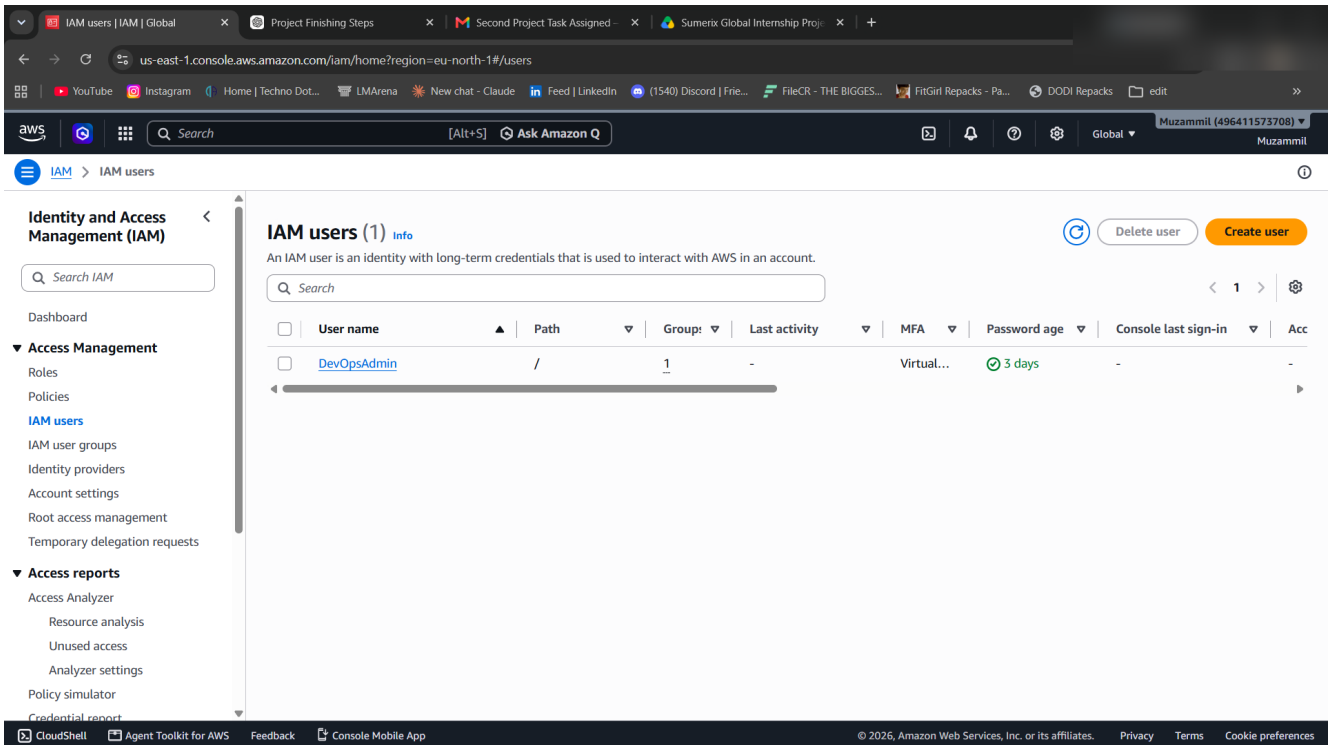
CloudWatch monitoring was configured for the EC2 infrastructure. Alarm: **ec2-CPU-High-70**; condition: EC2 CPU utilization greater than 70 percent.

7. Cost Estimation

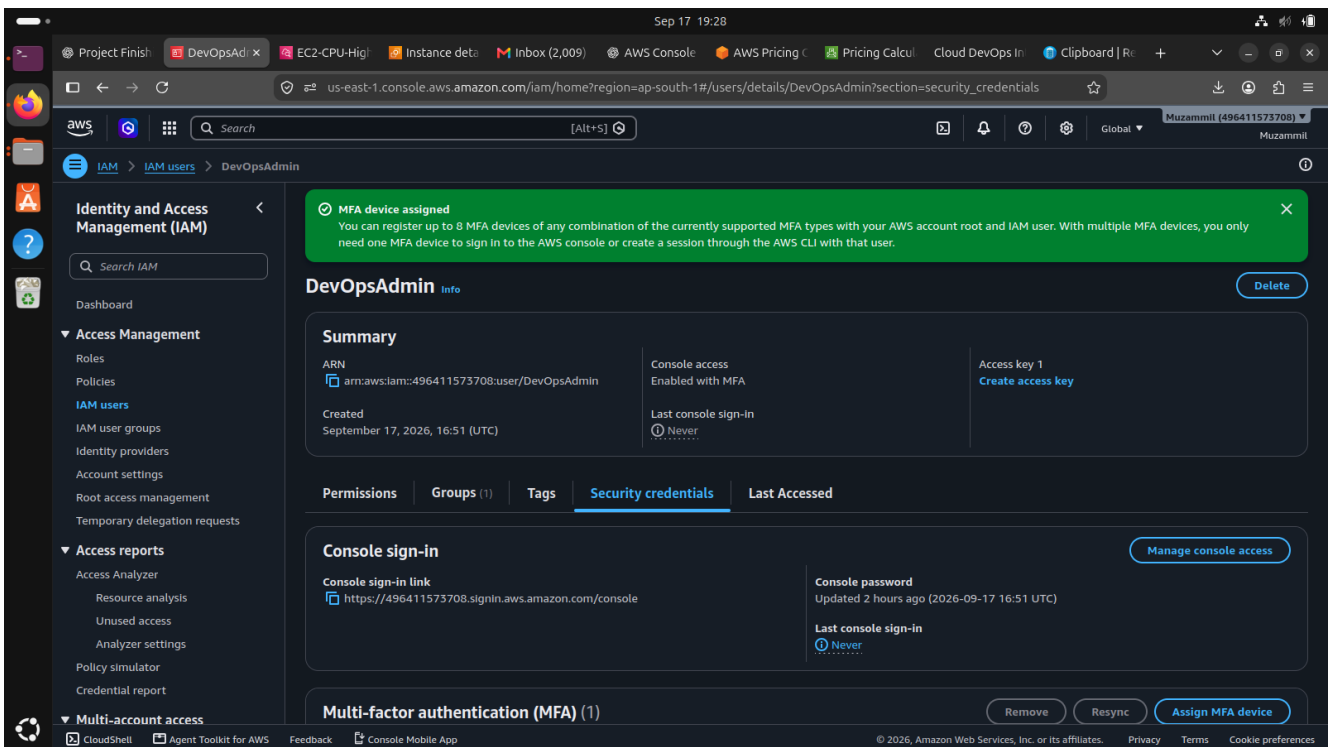
AWS Pricing Calculator evidence records an estimated monthly cost of **USD 15.84** and estimated 12-month cost of **USD 190.08**. Actual charges can vary with usage, data transfer, storage, taxes, pricing changes, and Free Tier eligibility.

8. IAM Evidence

IAM Users — DevOpsAdmin and CloudIntern



IAM user MFA evidence



CloudIntern MFA assignment

CloudIntern | IAM | Global

us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/users/details/CloudIntern?section=security_credentials

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

- Roles
- Policies
- IAM users**
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
- Resource analysis
- Unused access
- Analyzer settings
- Policy simulator
- Credential report

MFA device assigned

You can register up to 8 MFA devices of any combination of the currently supported MFA types with your AWS account root and IAM user. With multiple MFA devices, you only need one MFA device to sign in to the AWS console or create a session through the AWS CLI with that user.

CloudIntern Info

ARN: [arn:aws:iam::496411573708:user/CloudIntern](#)

Console access: Disabled

Access key 1: [Create access key](#)

Created: September 21, 2026, 15:18 (UTC+05:30)

Last console sign-in: -

Permissions: GROUPS (1) TAGS (0) **Security credentials** Last Accessed

Console sign-in

Console sign-in link: [https://496411573708.signin.aws.amazon.com/console](#)

Console password: Not enabled

Enable console access

Multi-factor authentication (MFA) (1)

Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA

Remove Resync Assign MFA device

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IAM Roles — EC2-S3-Access

Roles | IAM | Global

us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

- Roles**
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Roles (4) Info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Search

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Service-Linked Role)	23 minutes ago
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linked Role)	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linked Role)	-
☐	EC2-S3-Access	AWS Service: ec2	3 days ago

Roles Anywhere Info

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.

X.509 Standard

Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.

Temporary credentials

Use temporary credentials with ease and benefit from the enhanced security they provide.

Manage

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Root account MFA

Project Finishing Steps

Dashboard | IAM | Global

Inbox (2,009) - muzammil

us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/home

Search

Global

Muzammil (496411573708)

Muzammil

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

Access Analyzer

Resource analysis

Unused access

Analyzer settings

Policy simulator

Credential report

Multi-account access

IAM Dashboard

Info

Security recommendations

0

Root user has MFA

Having multi-factor authentication (MFA) for the root user improves security for this account.

Root user has no active access keys

Using access keys attached to an IAM user instead of the root user improves security.

IAM resources

Resources in this AWS Account

User groups

0

Users

0

Roles

4

Policies

0

Identity providers

0

What's new

Updates for features in IAM

AWS IAM now provides higher maximum quotas for roles, role trust policies, instance profiles, managed policies, and identity providers. 4 months ago

IAM Roles Anywhere now enforces VPC endpoint policies for the CreateSession API. 4 months ago

IAM Roles Anywhere now supports post-quantum digital certificates. 6 months ago

View all

AWS Account

Account ID

496411573708

Account Alias

Create

Sign-in URL for IAM users in this account

https://496411573708.signin.aws.amazon.com/console

Quick Links

My security credentials

Manage your access keys, multi-factor authentication (MFA) and other credentials.

Tools

Policy simulator

The simulator evaluates the policies that you choose and determines the effective permissions for each of the actions that you specify.

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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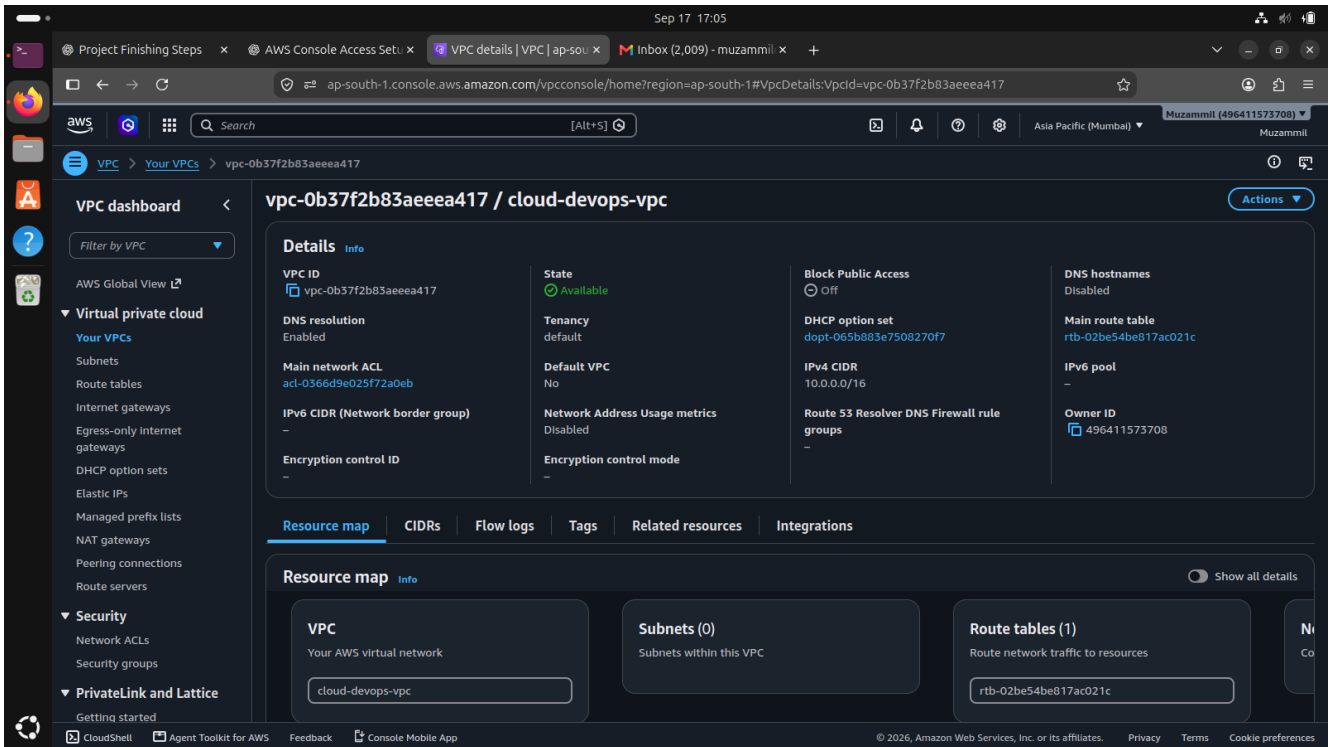
Privacy

Terms

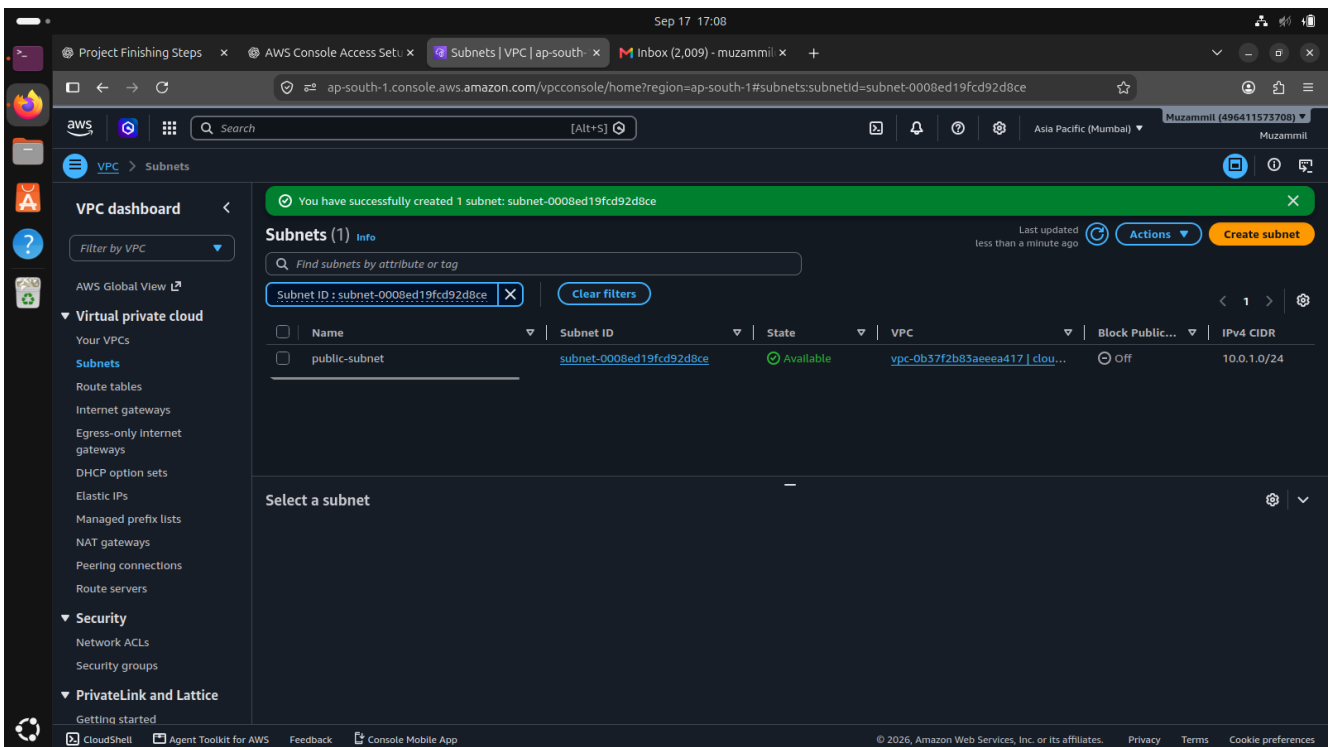
Cookie preferences

9. VPC and Networking Evidence

VPC



Public subnet



Private subnet

Sep 17 20:08

Project Finish x Subnet detail x Instance deta x Inbox (2,010) x AWS Console x AWS Pricing x Pricing Calcul x Cloud DevOps In x Clipboard | R x +

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SubnetDetails:subnetid=subnet-0b4eff9e6c5ae584f

Search [Alt+S]

Asia Pacific (Mumbai) Muzammil (496411573708) Muzammil

VPC dashboard < subnets > subnet-0b4eff9e6c5ae584f

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

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subnet-0b4eff9e6c5ae584f / private-subnet

Details

Subnet ID subnet-0b4eff9e6c5ae584f	Subnet ARN arn:aws:ec2:ap-south-1:496411573708:subnet/subnet-0b4eff9e6c5ae584f	State Available	Block Public Access Off
IPv4 CIDR 10.0.2.0/24	Available IPv4 addresses 251	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone aps1-az1 (ap-south-1a)	Network border group ap-south-1	VPC vpc-0b37f2b83ae417 cloud-devops-vpc	Route table rtb-0b40a233d36c0f0d6 private-route-table
Network ACL acl-0366d9e025f72a0eb	Default subnet No	Auto-assign public IPv4 address No	Auto-assign IPv6 address No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -	IPv4 CIDR reservations -
IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name	Resource name DNS A record Disabled
Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 496411573708	

Flow logs Route table Network ACL CIDR reservations Sharing Tags Related resources

Flow logs

Search

Actions Create flow log

< 1 >

Private route table

Sep 17 20:06

Project Finish x Route table x Instance deta x Inbox (2,010) x AWS Console x AWS Pricing x Pricing Calcul x Cloud DevOps In x Clipboard | R x +

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-0b40a233d36c0f0d6

Search [Alt+S]

Asia Pacific (Mumbai) Muzammil (496411573708) Muzammil

VPC dashboard < route tables > rtb-0b40a233d36c0f0d6

Filter by VPC

AWS Global View

Virtual private cloud

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Elastic IPs

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Peering connections

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You have successfully updated subnet associations for rtb-0b40a233d36c0f0d6 / private-route-table.

rtb-0b40a233d36c0f0d6 / private-route-table

Details Info

Route table ID rtb-0b40a233d36c0f0d6	Main No	Explicit subnet associations subnet-0b4eff9e6c5ae584f / private-subnet	Edge associations -
VPC vpc-0b37f2b83ae417 cloud-devops-vpc	Owner ID 496411573708		

Routes Subnet associations Edge associations Route propagation Tags Related resources

Explicit subnet associations (1)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
private-subnet	subnet-0b4eff9e6c5ae584f	10.0.2.0/24	-

Edit subnet associations

Subnets without explicit associations (0)

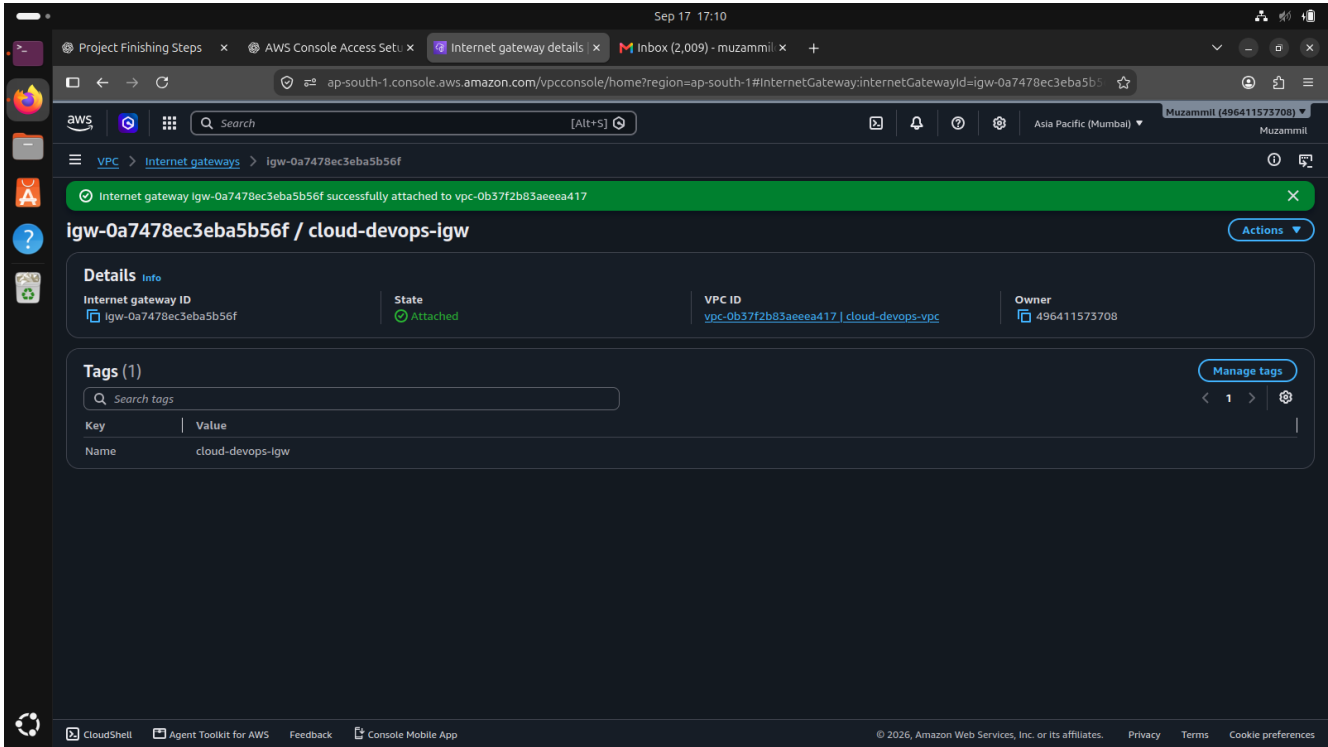
The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table.

Find subnet association

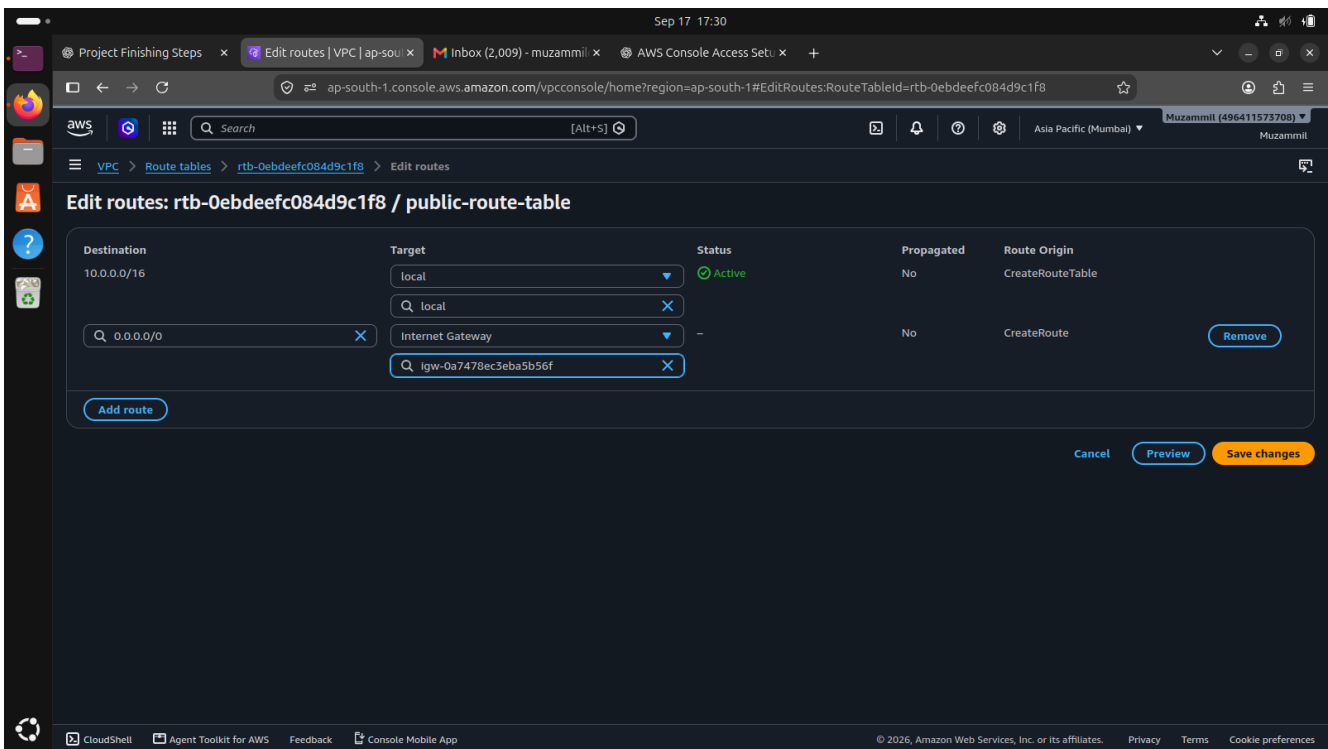
Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

Edit subnet associations

Internet Gateway

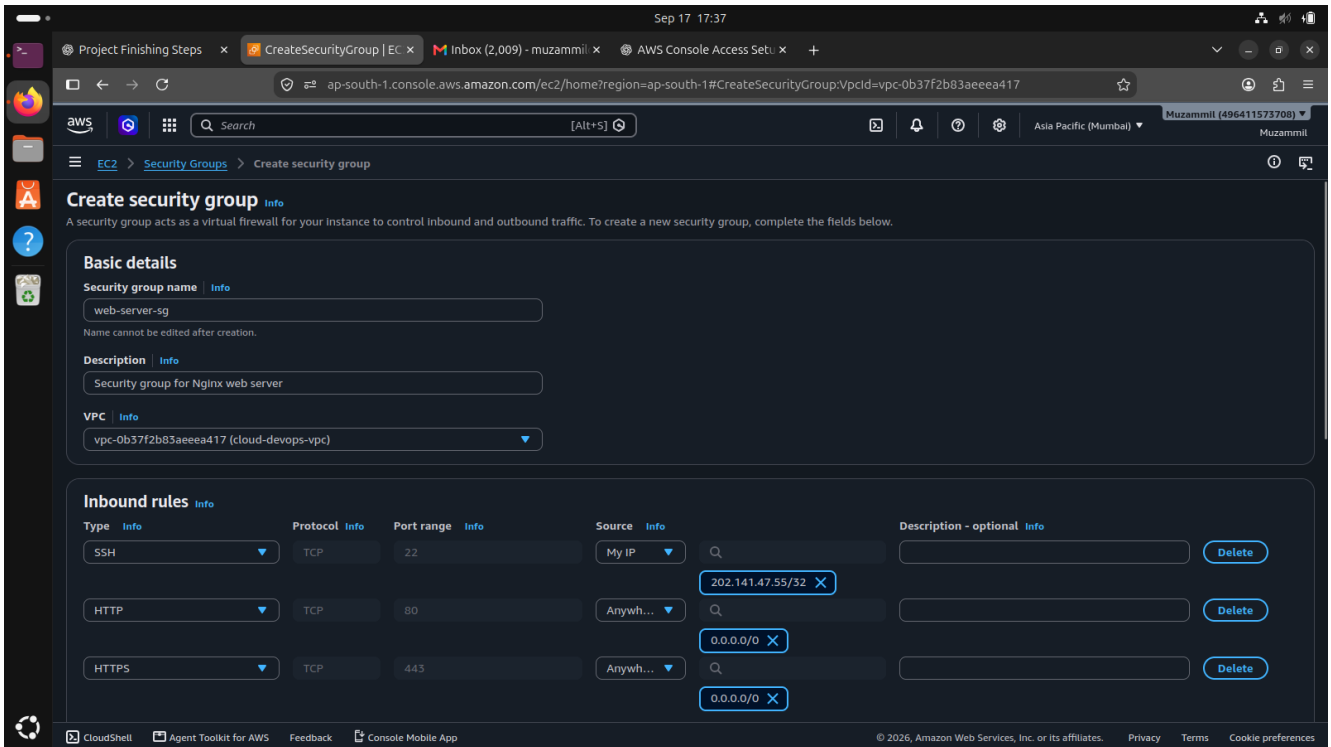


Route table

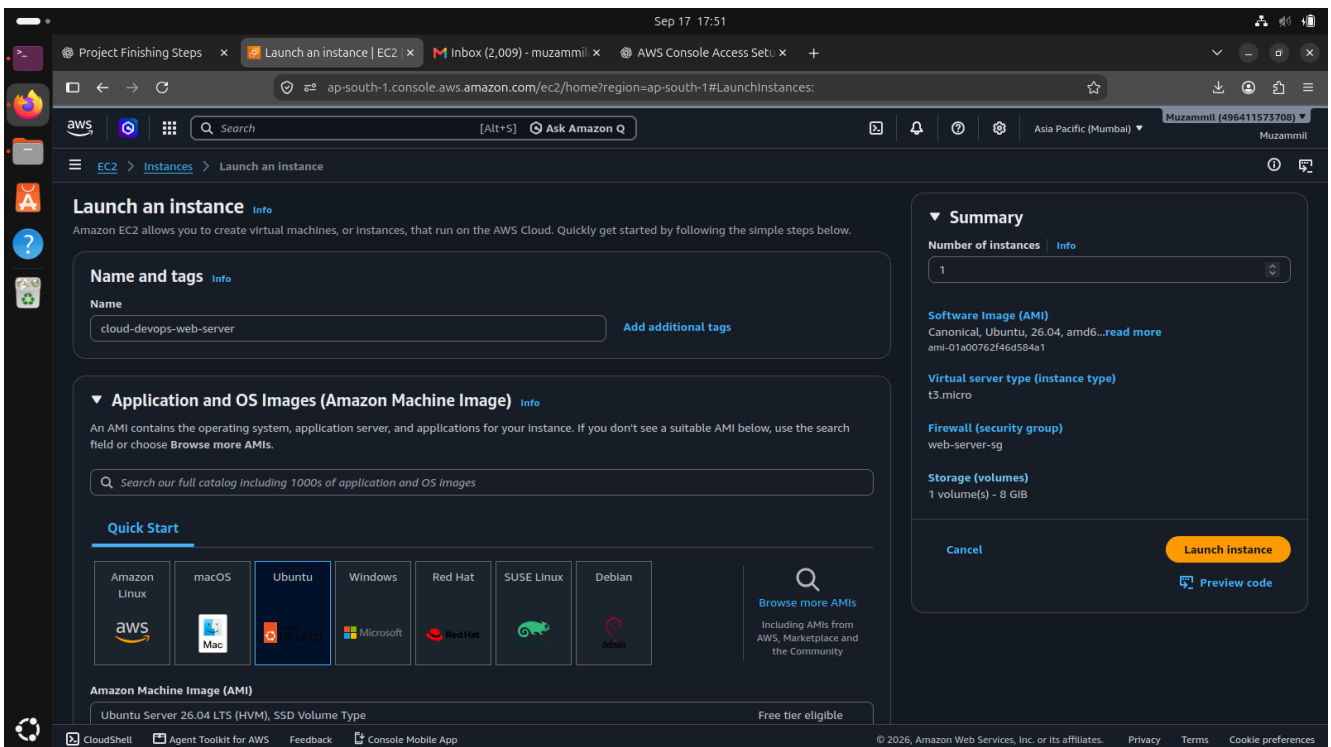


10. EC2, SSH and Nginx Evidence

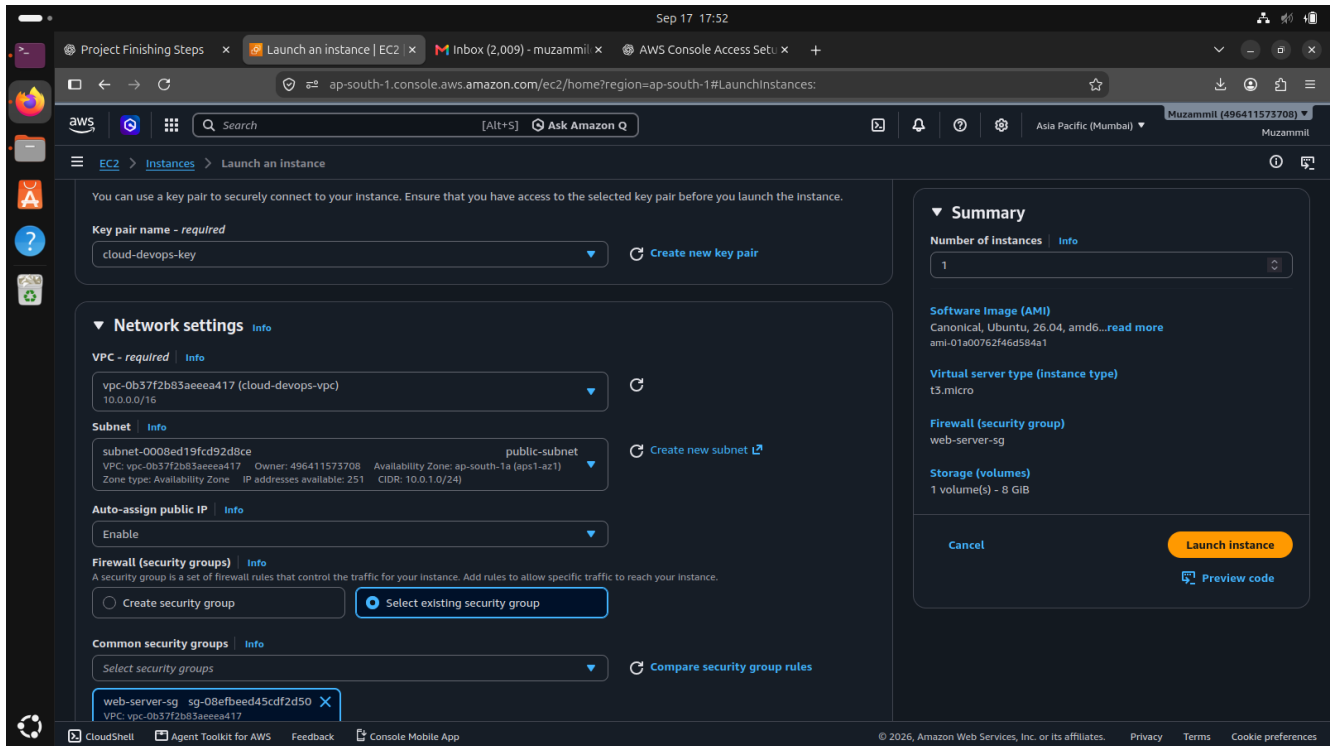
Security Group



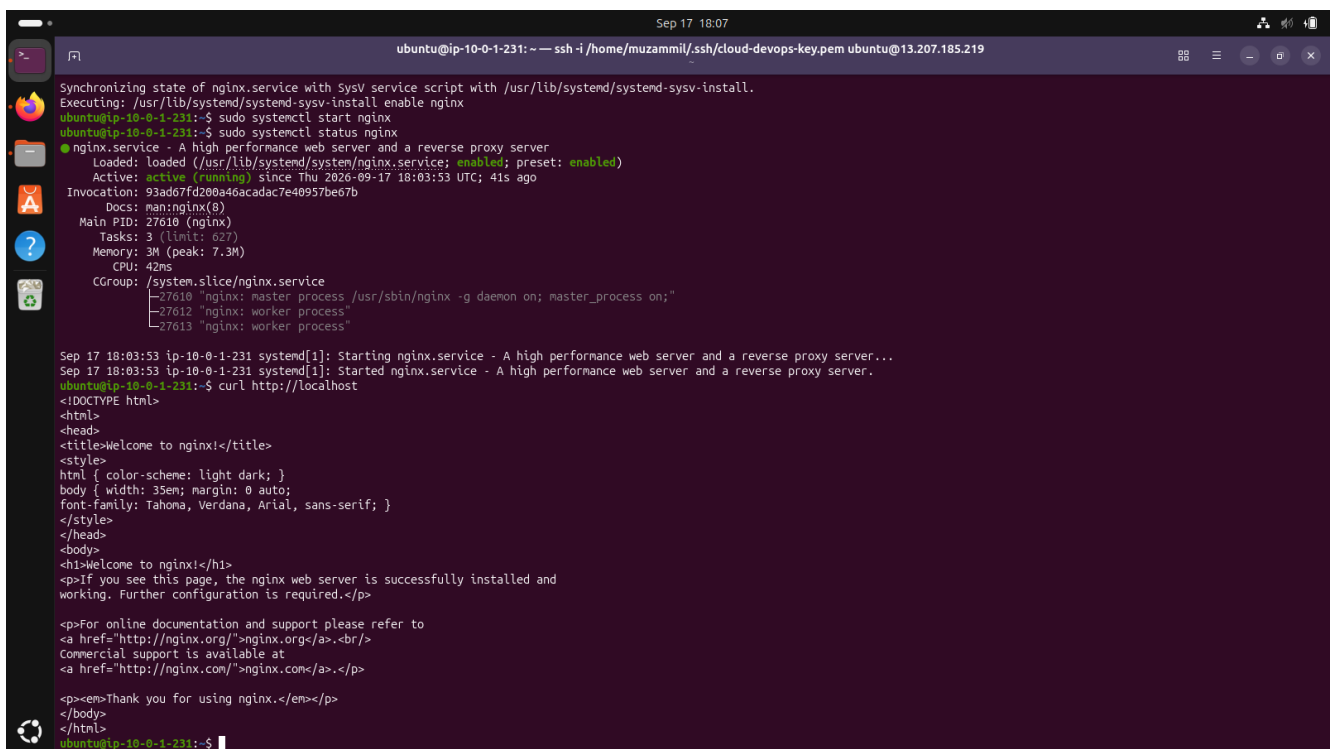
EC2 launch configuration



EC2 instance configuration



SSH connection



Nginx default page

```
Sep 17 18:14
ubuntu@ip-10-0-1-231: ~ -- ssh -i /home/muzammil/.ssh/cloud-devops-key.pem ubuntu@13.207.185.219

<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>

ubuntu@ip-10-0-1-231:~$ sudo nano /var/www/html/index.html
ubuntu@ip-10-0-1-231:~$ sudo systemctl restart nginx
ubuntu@ip-10-0-1-231:~$ Read from remote host 13.207.185.219: Connection reset by peer
Connection to 13.207.185.219 closed.
client_loop: send disconnect: Broken pipe
muzammil@student-server:~$ ssh -t ~/.ssh/cloud-devops-key.pem ubuntu@13.207.185.219
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1806-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 18:14:10 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   39.4% of 6.61GB Processes:      124
Memory usage: 33%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.1.231

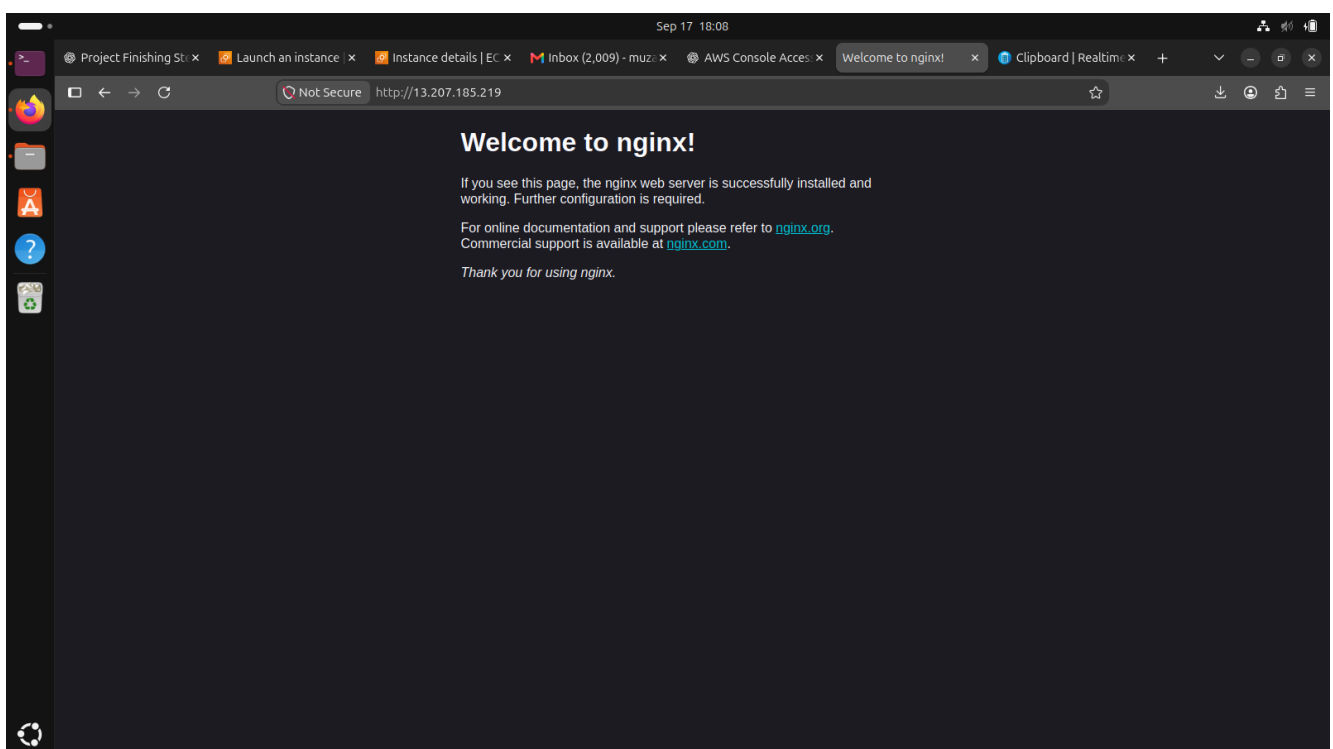
Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

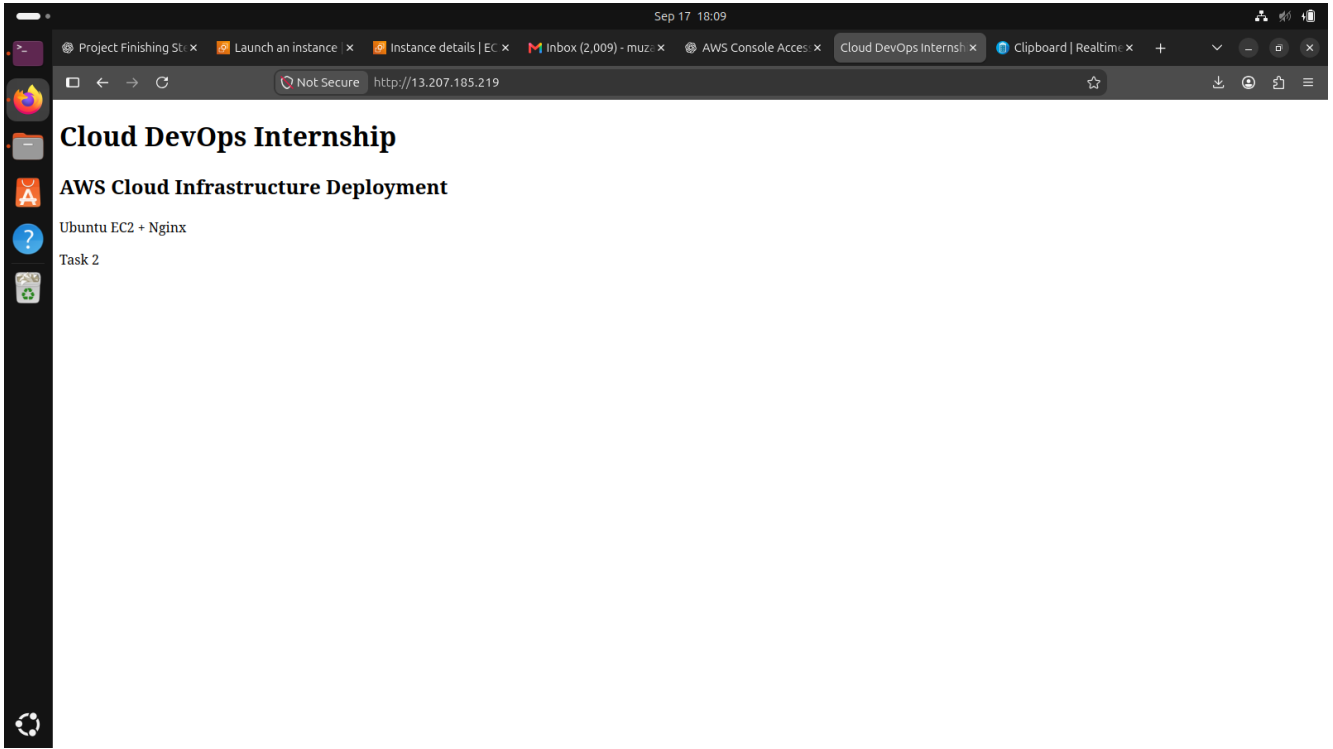
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Thu Sep 17 17:57:11 2026 from 202.141.47.55
ubuntu@ip-10-0-1-231:~$ echo "Cloud DevOps Internship - AWS S3 Task 2" > s3-test.txt
ubuntu@ip-10-0-1-231:~$
```

Nginx browser verification

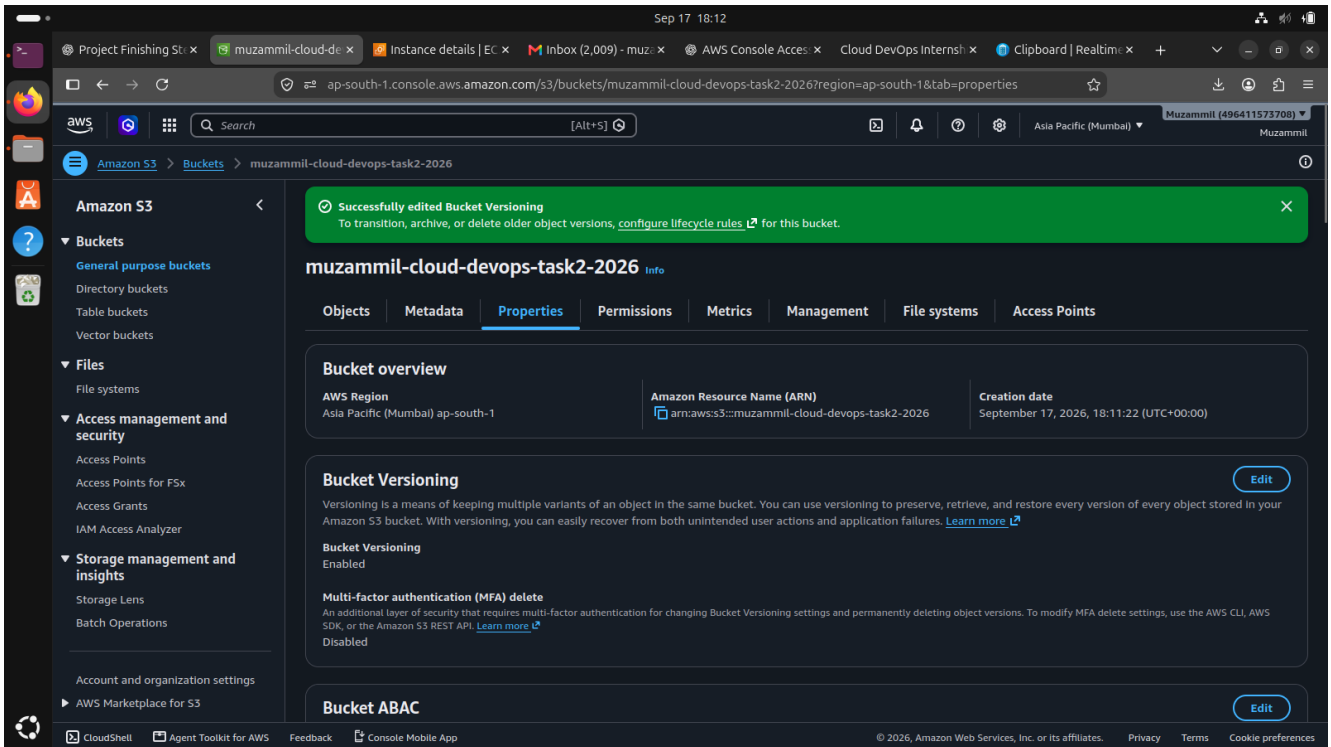


Custom Nginx page

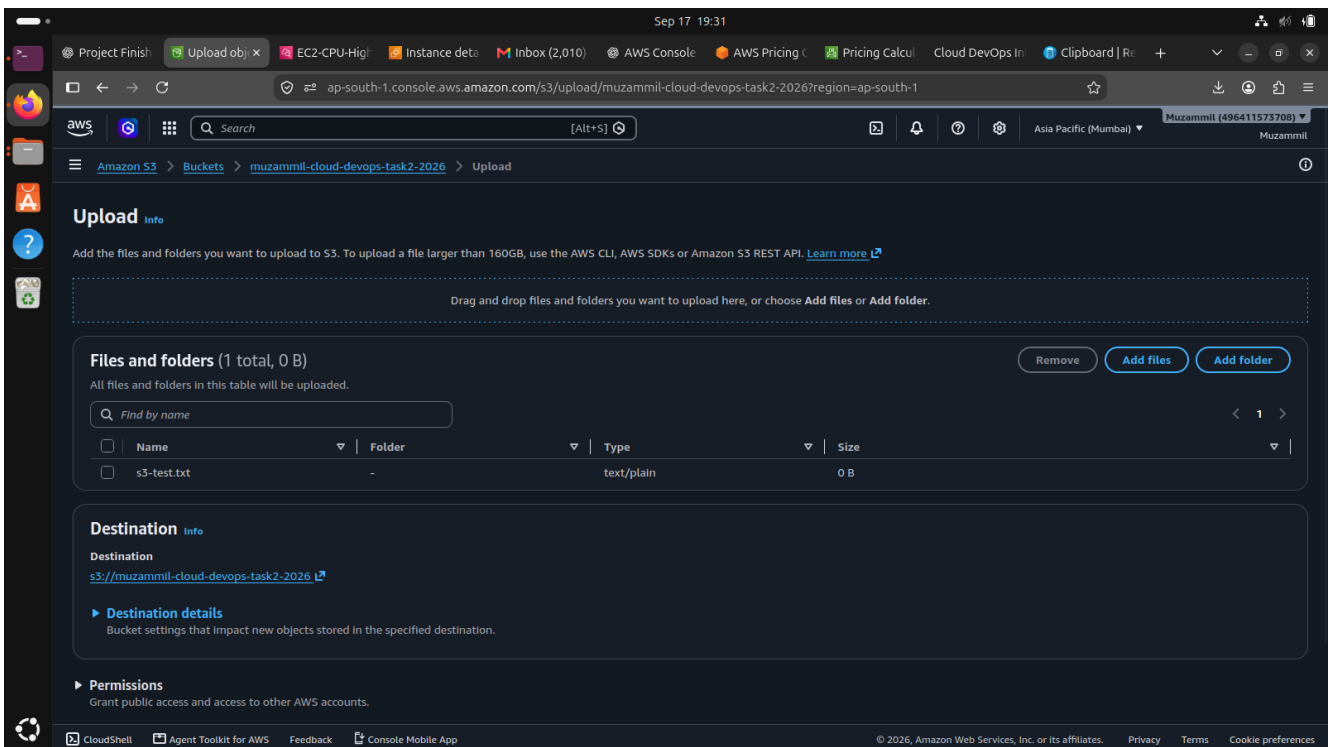


11. S3 Evidence

S3 bucket and versioning



S3 object upload



S3 uploaded object

Project Finish

Upload obj

EC2-CPU-High

Instance det

Inbox (2,010)

AWS Console

AWS Pricing

Pricing Calcul

Cloud DevOps In

Clipboard | Re

ap-south-1.console.aws.amazon.com/s3/upload/muzammil-cloud-devops-task2-2026?region=ap-south-1

Search [Alt+S]

Asia Pacific (Mumbai)

Muzammil (496411573708)

Muzammil

Amazon S3

Upload succeeded

For more information, see the [Files and folders](#) table.

Upload: status

Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination

s3://muzammil-cloud-devops-task2-2026

Succeeded

1 file, 0 B (0%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 0 B)

Find by name

Name	Folder	Type	Size	Status	Error
s3-test.txt	-	text/plain	0 B	Succeeded	-

CloudShell

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Feedback

Console Mobile App

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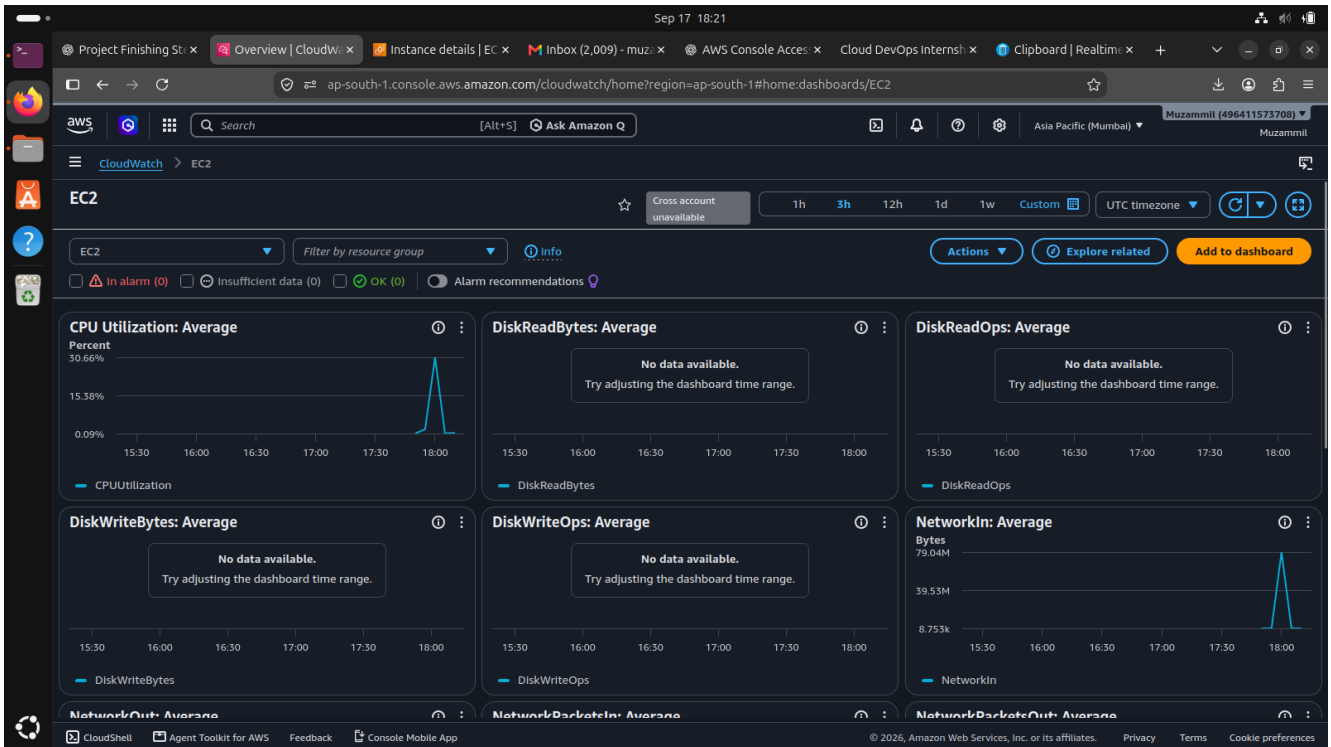
Privacy

Terms

Cookie preferences

12. CloudWatch, Architecture and Cost Evidence

CloudWatch dashboard



CloudWatch alarm configuration

The screenshot shows the 'Add alarm details' configuration page in the AWS CloudWatch console. The page is divided into four steps: Step 1 (Specify metric and conditions), Step 2 (Configure actions), Step 3 (Add alarm details), and Step 4 (Preview and create). The 'Add alarm details' step is currently active.

Add alarm details

Name and description

Alarm name: EC2-CPU-High-70

Alarm description - optional: [View formatting guidelines](#)

Edit **Preview**

Alarm when EC2 CPU utilization exceeds 70 percent.

Up to 1024 characters (50/1024).

[Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.](#)

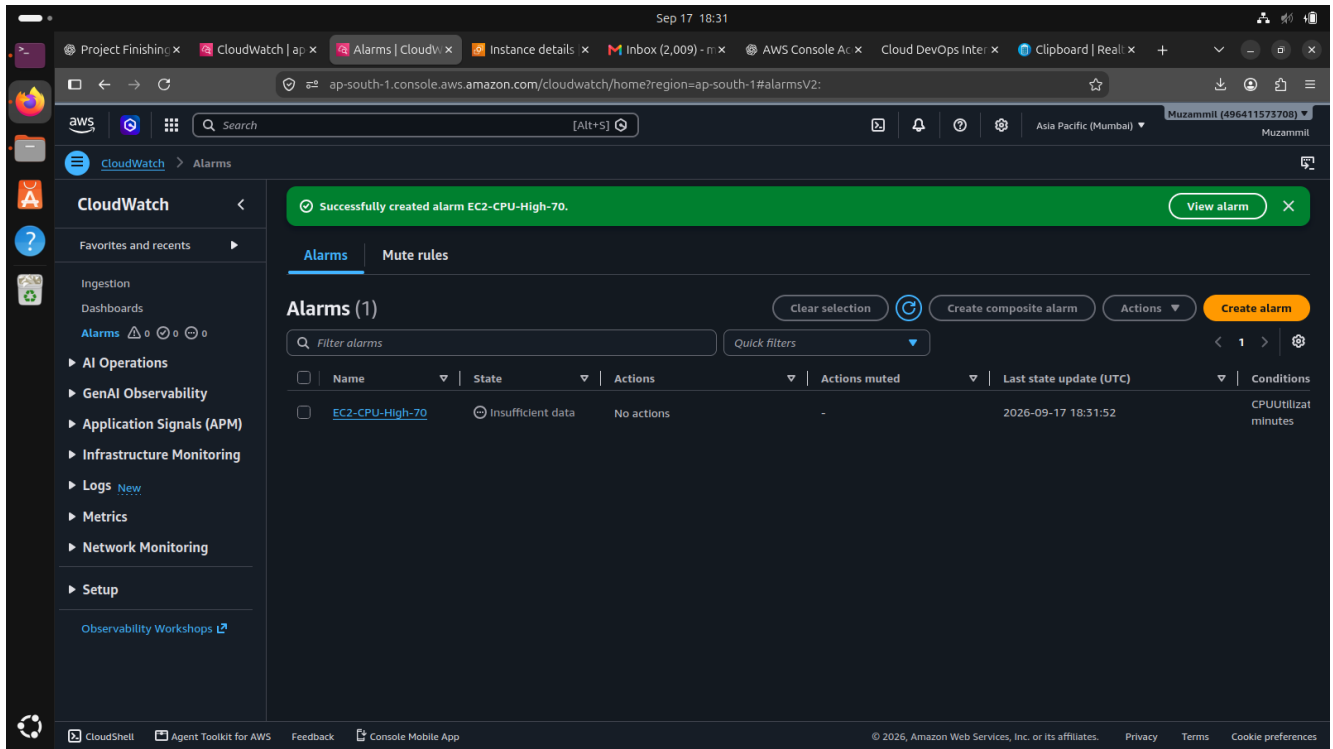
Tags - optional [Info](#)

No tags associated with the resource.

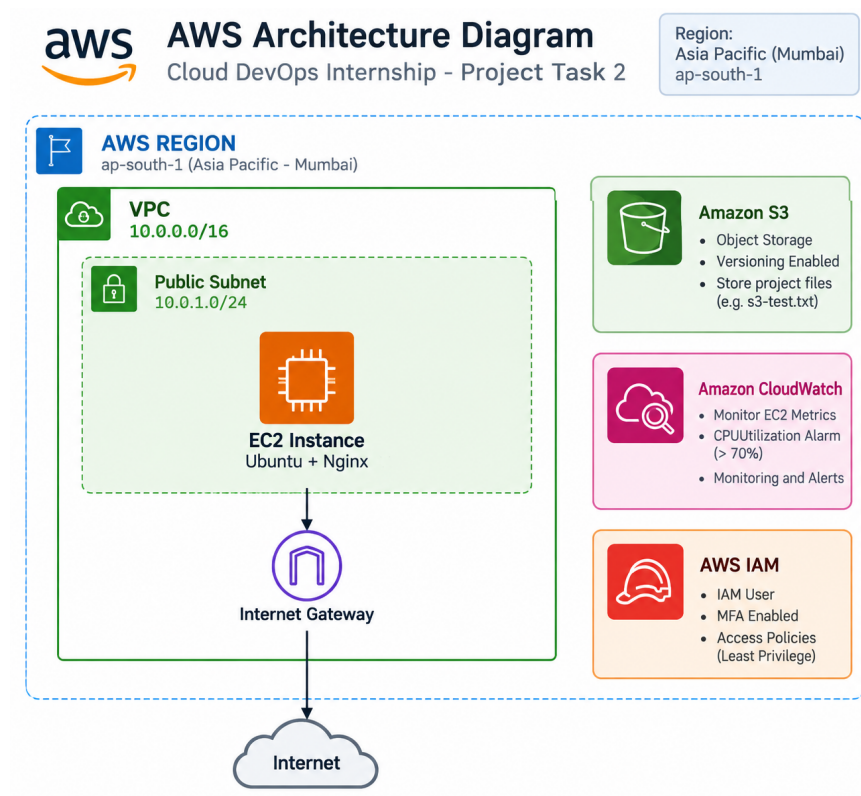
[Add new tag](#)

You can add up to 50 tags.

CloudWatch alarm created



AWS architecture diagram



AWS Pricing Calculator

Sep 17 19:54

Project FirMy Estir xPricing CaUpload obEC2-CPU-IInstance dInbox (2,0AWS ConsAWS PricirPricing CaCloud DevOpClipboard+

calculator.aws/#/estimate

aws pricing calculator

FeedbackLanguage: EnglishContact SalesCreate an AWS Account

Upfront cost
0.00 USD

Monthly cost
15.84 USD

Total 12 months cost
190.08 USD
Includes upfront cost

Get started for free

Contact Sales

My Estimate

DuplicateDeleteMove toCreate groupAdd supportAdd service

Find resources

☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region	Config Summary
☐	Amazon EC2	-	0.00 USD	15.84 USD	-	Asia Pacific (Mumbai)	Tenancy (Shared Insta...
☐	Amazon Simple Stora...	-	0.00 USD	0.00 USD	-	Asia Pacific (Mumbai)	
☐	Amazon CloudWatch	-	0.00 USD	0.00 USD	-	Asia Pacific (Mumbai)	Number of mobile OT...

Acknowledgement

AWS Pricing Calculator provides only an estimate of your AWS fees and doesn't include any taxes that might apply. Your actual fees depend on a variety of factors, including your actual usage of AWS services. [Learn more](#)

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13. Final Submission Status

The repository contains the required Task 2 implementation evidence: IAM users/groups/MFA/role evidence, custom VPC with public and private subnets, Internet Gateway and route tables, Security Group, Ubuntu EC2 with SSH, Nginx, S3 with versioning and object evidence, CloudWatch dashboard and alarm, architecture diagram, Pricing Calculator evidence, cost documentation, logs, README, screenshots, and this report.